Exploring the Limits of Machine-Learning Classification of Neutron-Star Matter Models:

Populations of Equations of State, Illustrative Observational Errors and Calibrated Uncertainty

Wasif Husain
School of Science, Technology and Engineering, University of Sunshine Coast, Adelaide 5000

Revised manuscript (supersedes arXiv:2512.23758v3)

Abstract

Machine-learning classification of neutron-star composition is usually demonstrated on synthetic data in which every composition is a single equation of state (EoS) and test points are random draws along the same sequences. We show first that the headline 97% accuracy of a released benchmark of 770 such models is largely interpolation between neighbouring models (the MLP accuracy falls from 97% to 94% and 69% when blocks of six and 24 models are withheld) and that its tabulated surface redshift leaks class information. We then replace the benchmark by a population study of 360 EoS, 90 in each of four composition scenarios (nucleonic, hyperonic-like, strange, and neutron-decay dark-matter-admixed), solved with a full general-relativistic TOV, tidal and f-mode code written for this work. The dark-matter family uses only dark-fermion masses inside the stability window ($m_\chi \leq 938.783\,\mathrm{MeV}$); a wider prior is a sensitivity run. The code agrees to 0.007% with six f-mode frequencies from an unpublished same-author pilot (computed with a third-party code) that shares the EoS inputs (a code-to-code consistency check, not an absolute accuracy), with analytic benchmarks (the Kelvin mode and Love number of an incompressible star, and the Schwarzschild quasi-normal mode), and gives damping times 2.9–7.0% above a published universal relation for nuclear EoS. With whole-EoS hold-out, six classifiers (logistic regression, k-NN, random forest, gradient boosting, SVM and an MLP ensemble) reach only 65% accuracy even without measurement error (95% bootstrap interval 61–67%, which resamples the EoS of the fixed catalogue and does not include regenerating the populations or retraining); the tested learners and a finite-reference kernel-Bayes classifier that uses the exact error model perform similarly. Strange matter is separable (100% recall when f is measured accurately), hyperonic-like matter partly (76%), but no discrimination between nucleonic and dark-matter-admixed matter is demonstrated in these experiments once the nuclear EoS is unknown (pairwise accuracy 51%, interval 45–56%, which includes chance); with the nuclear EoS known exactly the dark component shifts f by a median of 0.57% and would need an f-mode precision of about 0.51% (interval 0.45–0.59%) to be detected at 75%. This shift depends on how baryon number is assigned to the one-fluid barotrope: changing the normalisation m_0 to 938.9 MeV changes the shift of an individual EoS by a median of 5% and at most 6% inside the stability window, but by up to 65% for dark-fermion masses above it. With illustrative errors taken from different published measurements and forecasts (NICER-like 10% in M and 9% in R, and a 20% f-mode measurement, which are not a realistic joint observation of one star) the accuracy is 33–37% (chance 25%) and conformal prediction sets contain 3.22 of four classes; 5%, 5% and 2% errors would give 53%, and the damping time adds nothing. Probabilities are well calibrated, and conformal prediction sets built from one star per calibration EoS have empirical marginal coverage of 0.91–0.92 at the 0.90 target for stars of new EoS drawn from the same catalogue; this is an empirical result, not a distribution-free guarantee, because the folds are stratified and class-balanced. Repeating the analysis on an alternative population (different parametrisations and priors) again shows no demonstrated nucleonic–dark-matter discrimination (51%) and a similar NICER-like accuracy (37%), whereas the 65% plateau and the identifiability of the hyperonic-like family depend on how that family is built. This is a controlled identifiability and forecasting study: all results are

conditional on the phenomenological EoS families, priors and error models defined here, and no inference about real stars is claimed.

1 Introduction

Neutron stars probe matter at densities well above nuclear saturation (Lattimer & Prakash, 2004; Özel & Freire, 2016), and a wide range of compositions—purely nucleonic matter, hyperon-rich matter, deconfined quark matter and neutron decay into a dark sector (Husain et al., 2022)—has been proposed. Machine learning has been applied to neutron-star inference, for example to map observed masses and radii to the EoS (Fujimoto et al., 2018). A natural question is how far a supervised classifier can distinguish composition from observables, and where the answer is limited by physics rather than by the algorithm.

A previous version of this work (arXiv:2512.23758v3) trained a small multilayer perceptron (MLP) on seven observables of 770 synthetic stellar models, each class being one EoS, and reported a test accuracy of 97.4% on 116 held-out models. That number does not support the conclusions drawn from it, and a single sequence per class cannot say how well compositions can be told apart when the EoS itself is uncertain. This revision therefore does the following.

1. **Audits the single-sequence benchmark** (Sec. 2): class sizes, mass coverage, a class-dependent redshift inconsistency, and the dependence of the accuracy on how test points are chosen.

2. **Builds a population of EoS** for each composition scenario and solves every one with a validated full-GR code (Sec. 3); the checks (agreement with the six f-mode frequencies of an unpublished same-author cross-code pilot, analytic benchmarks and a published damping-time relation) are described in Sec. 3.2 and Table 4.

3. **Evaluates with whole-EoS hold-out**, six learners and a kernel-Bayes reference classifier (Secs. 4, 5), under error scenarios taken from published measurements and forecasts as well as a scale that shows what precision would be needed.

4. **Quantifies uncertainty**: calibration, temperature scaling and conformal prediction sets with exchangeability at the level of whole EoS.

Appendix A keeps the single-sequence noise study, and Appendix C lists each claim of the earlier version that has been withdrawn or changed.

2 What the single-sequence benchmark can and cannot show

2.1 Audit of the released data set

The released data (`DataML.xlsx`) hold 770 models of four classes: a nucleonic and a hyperonic quark–meson-coupling EoS (Rikovska Stone et al., 2007), a neutron-decay dark-matter scenario (Husain et al., 2022) and MIT-bag strange matter; the oscillation quantities come from the earlier oscillation code of Husain (2026a) and are used as released here. Table 1 and Fig. 1 summarise the data.

1. *Class sizes.* The classes contain 222, 220, 122 and 206 models (not ≈ 190 each).

2. *Sampling.* Models lie on a regular radius grid ($\Delta R = 0.05\,\mathrm{km}$) along each sequence, so neighbouring models are almost identical: for 96.5% of the models the nearest neighbour in standardised $(\ln M, \ln R, \ln f, \ln \tau)$ belongs to the same class.

3. *Mass coverage.* 49% of the models have $M < 0.5\,M_\odot$ and 70% have $M < 1\,M_\odot$, down to $7.2 \times 10^{-8}\,M_\odot$.

Table 1: Composition of the released single-sequence data set (770 models). $N_{\geq 0.5}$ and $N_{\geq 1}$ are the numbers of models with $M \geq 0.5\,M_\odot$ and $M \geq 1\,M_\odot$.

Class	N	M range ($M_\odot$)	R range (km)	$M < 0.5\,M_\odot$	$N_{\geq 0.5}$	$N_{\geq 1}$
Nucleonic	222	7.2×10^{-8}–2.16	0.06–11.11	47%	118	83
Strange	220	4.2×10^{-6}–1.94	0.11–11.06	45%	122	80
Dark-matter	122	1.7×10^{-3}–0.68	0.51–6.56	67%	40	0
Hyperonic	206	2.0×10^{-4}–1.74	0.31–10.56	44%	115	69

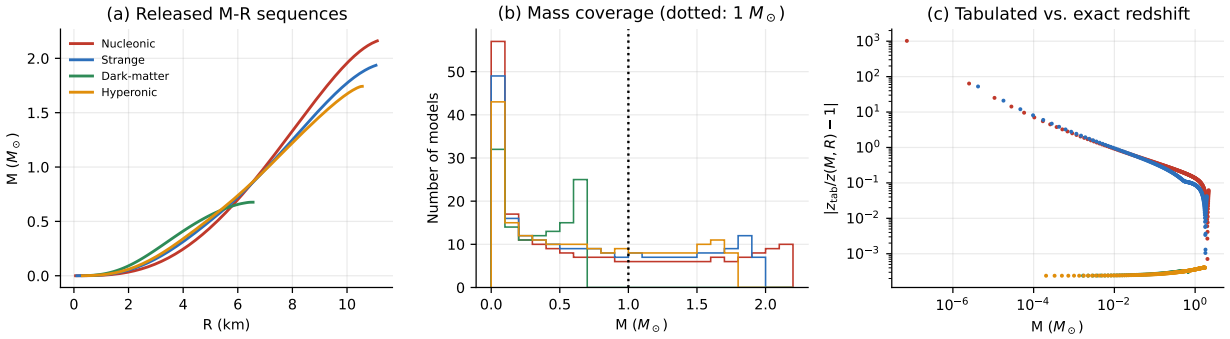


Figure 1: (a) Mass–radius sequences of the four classes. (b) Distribution of masses; the dotted line is $1\,M_\odot$. (c) Fractional difference between the tabulated surface redshift and the value implied by M and R; the hyperonic and dark-matter classes agree to $\sim$0.03%, the nucleonic and strange classes do not.

4. *The dark-matter sequence ends at* $0.68\,M_\odot$, so at $M \gtrsim 0.7\,M_\odot$ the class is excluded by the mass alone.

5. *Redshift inconsistency.* For the dark-matter and hyperonic classes the tabulated z equals the Schwarzschild value implied by (M, R) to $\approx$0.031%; for the nucleonic and strange classes the median deviation is +18% and +11%. A logistic regression on the redshift residual alone separates {nucleonic, strange} from {dark matter, hyperonic} with 91% cross-validated accuracy (57% for M alone, 72% for M and R). The redshift is therefore a class-dependent artefact and is not used in any analysis here.

2.2 Replicating the earlier protocol

Table 2 and Fig. 2 repeat the earlier random-split protocol for 100 seeds. The published 97.4% is typical (97.7 ± 1.6%), so it is reproducible, but it is not specific to the MLP: a random forest reaches 99.2%. Removing the redshift changes the MLP result only from 97.7% to 97.5%. With M and R only, a k-nearest-neighbour classifier reaches 60.1%, and with M alone logistic regression is at chance (25.3%).

2.3 Leakage from neighbouring models

Figure 3 holds out contiguous blocks of increasing length (noise-free, 70/30 split, 15 repetitions). With all six observables (M, R, f, Q, τ, h_c) the MLP accuracy decreases from 97% (single-model holdout, equivalent to a random split) to 94% at six-model blocks and 69% at 24; the random forest drops from 99% to 82% and 42%. With the four plausible observables M, R, f, τ the MLP values are 78%, 69% and 45%. A large part of the apparent skill is therefore interpolation between near-duplicate neighbours of the same sequence. Q and h_c are model-derived amplitudes rather than measurable quantities and are not used further. Even with the leakage removed, a single sequence per class cannot say whether the *composition* is identifiable when the EoS is uncertain: "classifying the family" and "recognising one EoS" are the same task. The remainder of the paper addresses this with populations.

Table 2: Random-split test accuracy (%) over 100 seeds (stratified 70/15/15, $N_{\text{test}} = 116$), noise-free.

Setup	Mean	s.d.	Min–max
Published setup: 7 columns incl. z, MLP	97.7	1.6	92.2–100.0
Published setup, random forest	99.2	1.1	94.0–100.0
Redshift removed, MLP	97.5	1.9	90.5–100.0
Redshift removed, random forest	98.9	1.2	94.0–100.0
Redshift removed, k-NN	98.7	1.0	96.6–100.0
M, R, f, τ only, MLP	93.6	3.5	79.3–99.1
M, R only, k-NN	60.1	4.1	50.9–69.8
M, R only, logistic regression	27.1	4.0	19.0–41.4
M only, logistic regression	25.3	3.4	15.5–31.9

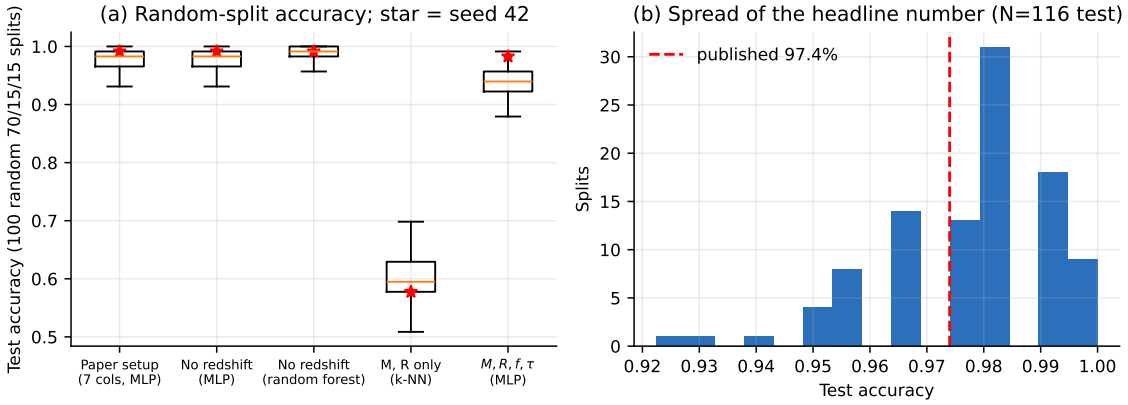


Figure 2: (a) Test accuracy over 100 random splits for several setups (star: seed 42, used in the earlier version). (b) Distribution for the original setup.

3 A population of equations of state

3.1 Composition scenarios as EoS populations

Each scenario is a family of EoS defined by a small number of parameters with a uniform prior (Table 3); one draw from the prior is one EoS. The families are phenomenological parametrisations *inspired by* the four scenarios of the earlier version, not microphysical calculations. The dark-fermion mass window follows the neutron-decay dark-matter scenario (Fornal & Grinstein, 2018): $m_\chi > 937.9\,\text{MeV}$ is required for the stability of ^{9}Be and $m_\chi < m_p + m_e = 938.783\,\text{MeV}$ for the stability of χ. Both bounds are imposed in the primary population, so every dark-matter EoS has m_χ inside the window. An earlier version of this analysis drew m_χ up to 939.2 MeV, which exceeds the second bound for 27 of its 90 dark-matter EoS; that population is kept as a sensitivity run (Sec. 5.7). Draws are accepted only if they satisfy broad observational and consistency screens: $M_{\max} \geq 1.97\,M_\odot$, a causal speed of sound ($c_s^2 \leq 1$) up to the centre of the maximum-mass star, $70 \leq \Lambda_{1.4} \leq 800$ and $10 \leq R_{1.4} \leq 14\,\text{km}$. The acceptance rates are 21%, 57%, 21% and 26% for the nucleonic, strange, dark-matter and hyperonic-like priors, so the screens remove most of the prior volume (mainly through $M_{\max}$ and causality, and for strange matter through $\Lambda_{1.4}$ and $R_{1.4}$). Observables are tabulated for stars of 1.2–1.9 $M_\odot$ and evaluated on a mass grid over 1.2–1.85 $M_\odot$, where all accepted EoS have stable stars.

The dark-matter-admixed scenario follows the effective-barotrope construction of the earlier work: baryons (nucleonic PP, $\mu_N = (\varepsilon + p)/n_B$) and a dark Fermi gas with $\mu_\chi = \sqrt{m_\chi^2 + k_F^2} + \hbar c G n_\chi$ share one chemical potential, and the total pressure and energy density are summed at that potential. Because the dark component appears only above $\mu_N = m_\chi$, the stellar surface is purely baryonic and the dark component is at most a few per cent of the energy density even at the highest densities. By construction this scenario shares the nuclear prior with the nucleonic one, so the question the

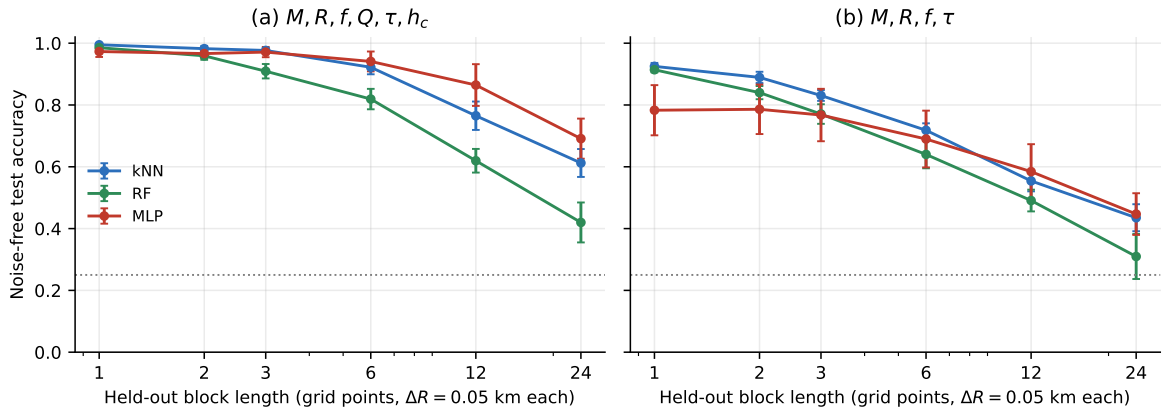


Figure 3: Noise-free test accuracy versus the length of the held-out contiguous mass block (mean and 95% t-interval over 15 repetitions). Dotted: chance. A block length of one is equivalent to the random split of the earlier version.

Table 3: Parametrisations and priors. PP: piecewise polytrope (Read et al., 2009) with core break densities $\rho_1 = 10^{14.7}$ and $\rho_2 = 10^{15}\,\mathrm{g\,cm^{-3}}$ on a four-piece crust; CSS: constant speed of sound, $\varepsilon = \varepsilon_0 + p/c_s^2$.

Scenario	Parameters (uniform priors)
Nucleonic	PP: $\log_{10} P_1 \in [34.15, 34.60]$ ($\mathrm{dyn\,cm^{-2}}$ at ρ_1), $\Gamma_1 \in [2.4, 3.8]$, $\Gamma_2 \in [2.4, 3.9]$, $\Gamma_3 \in [2.2, 3.9]$
Hyperonic-like	PP with $\log_{10} P_1 \in [34.20, 34.60]$, $\Gamma_1 \in [2.6, 3.8]$ and a soft segment $\Gamma_2 \in [1.8, 2.6]$ above $\rho_1 \approx 2n_0$ (hyperon onset), then a causal CSS segment with $c_s^2 \in [0.5, 1]$ above $\rho_t = 10^u$, $u \in [14.95, 15.25]$
Strange (self-bound)	CSS with $\varepsilon_0 = 4B$, $B^{1/4} \in [130, 160]$ MeV, $c_s^2 \in [1/3, 1]$ ($c_s^2 = 1/3$ is the MIT bag model)
Dark-matter admixed	nucleonic PP prior plus a self-interacting dark Fermi gas in chemical equilibrium with the baryons (neutron-decay scenario): $m_\chi \in [937.9, 938.783]$ MeV (stability window), repulsive self-interaction $G \in [50, 500]\,\mathrm{fm^2}$; effective one-fluid barotrope

classifier faces is whether the dark component can be recognised against the spread of nuclear EoS.

3.2 Solvers and validation

TOV and tidal deformability. Stars are integrated with the Tolman–Oppenheimer–Volkoff equations (Tolman, 1939; Oppenheimer & Volkoff, 1939) and the tidal-deformability equation (Hinderer, 2008).

Full-GR f-mode. The $l = 2$ quasi-normal f-mode of the non-rotating, barotropic star is computed from the linearised Einstein equations in the Regge–Wheeler gauge. The interior equations were derived symbolically (not taken from the literature) from the linearised Einstein equations and the conservation law, reduced to four first-order equations, and solved with regular boundary conditions at the centre and a vanishing Lagrangian pressure perturbation at the surface; this is the same physical system as the Lindblom–Detweiler formulation (Detweiler & Lindblom, 1985). The exterior solution is matched to the Zerilli equation (Zerilli, 1970), and the complex eigenfrequency is the zero of the ingoing-wave amplitude; the real part gives f and the imaginary part τ. The exterior equations were checked against an independent symbolic derivation (residual 10^{-12}). The complex eigenfrequency is found by a secant search on the analytic ingoing amplitude, starting within 3% of the root.

Numerical error budget. Varying one numerical setting at a time (interior and exterior integration tolerances from 10^{-8} to 10^{-13}, the TOV tolerance, the surface cut-off, the starting

radius at the centre and the outer matching radius from 1500 to 8000 km) changes f by at most 3e-08 in relative terms and τ by at most 5e-07, for SLy and for SLy with $G = 100\,\mathrm{fm}^2$ at $1.4\,M_\odot$. These are about seven orders of magnitude smaller than the dark-matter shift of interest (median 0.57%). The remaining sensitivity is physical: changing the crust pressure scale by ±10% shifts f of the nuclear EoS by 0.20% but changes the dark-matter shift only from 1.313% to 1.338% (SLy, $G = 100\,\mathrm{fm}^2$), so the *shift* is robust to the crust treatment even though absolute frequencies are not.

Damping times. Damping times are compared with a published relation in the paragraph on published damping times below. The time dependence is $e^{i\omega t}$, so $\tau = 1/\mathrm{Im}\,\omega$ and a damped mode has $\mathrm{Im}\,\omega > 0$. The exterior machinery that produces $\mathrm{Im}\,\omega$ (Zerilli integration, projection onto asymptotic ingoing and outgoing solutions, complex root search) was benchmarked on a case with a published answer: with the horizon matched to purely ingoing waves it gives the fundamental $l = 2$ quasi-normal mode of a Schwarzschild black hole, $M\omega = 0.37366 + 0.08896\,i$, against $0.37367 + 0.08896\,i$ (Leaver, 1985), i.e. relative errors of 3e-05 (real part) and 8e-06 (imaginary part). The interior contribution to τ at stellar compactness is tested only against a published fitting relation (below), with an offset of several per cent that is not understood, so statements about τ (in particular that it adds no classification information) are conditional on the solver.

Adiabatic response of the dark-matter configurations. The perturbation equations use the equilibrium barotrope, i.e. chemical equilibrium is assumed to be maintained during the oscillation. As a bound on the opposite limit, f was recomputed with a frozen-composition adiabatic response (baryon and dark numbers separately conserved, both components compressed together) for 24 dark-matter EoS at $1.4\,M_\odot$: the median dark-matter shift is 0.64% with the equilibrium response and 0.64% with the frozen one, a difference of 1.1% of the shift (at most 0.025% of f). The adiabatic response of the dark sector therefore does not matter at the level of the effect. This does not test independent displacements of the two fluids, nor the equilibrium assumption for the nuclear composition itself, which affects the absolute frequencies of all classes alike.

Validation. The crust is a single calibrated parameter (the absolute pressure scale of a four-piece SLy-like crust, 1.135 times the electron-gas value), fitted only to the radius and tidal deformability of six published EoS fits at $1.4\,M_\odot$ (Table 4); the radii are then reproduced to 0.0006 km and Λ to 0.05%. The f-mode frequencies were *not* used in any fit; they agree with the pilot's to 0.007% (maximum over the six EoS; ours are uniformly slightly higher). The reference values are those of the finite-EoS pilot of the earlier work (Husain 2026b), which used the separately developed RelModPy full-GR code (`https://github.com/fgittins/RelModPy`); the six values are listed in Table 4 so that the comparison can be repeated with any other code. The pilot is an unpublished working manuscript by the same author (its numbers were computed with the third-party RelModPy code, but they have not been independently checked or published) and shares with the present work the Read et al. polytropic fits, the six reference EoS and the effective-barotrope construction of the dark-matter scenario; only the f-mode solver is a different implementation. This agreement is therefore a code-to-code consistency check and is *not* an absolute accuracy of 0.007% for stellar f-modes: the reference values come from the unpublished same-author pilot, and its inputs (EoS, crust, barotropic perturbation model) are shared, so errors common to both codes would not be detected. What the comparison shows is that two implementations solve the same linearised problem for the same barotropes. The independent tests of the solver are the analytic benchmarks and the comparison with a published relation below; no comparison with published numerical tables of f-modes for realistic EoS has been made. In addition, the dark-matter frequency shift at fixed nuclear EoS (SLy, $G = 100\,\mathrm{fm}^2$, $1.4\,M_\odot$) is 25.7 Hz against 25.8 Hz in the reference, and a tabulated version of the same EoS agrees with the analytic one to 0.013% in f. Damping times of 0.17–0.23 s are obtained for the six EoS, as expected for a $1.4\,M_\odot$ f-mode; the pilot reference values do not include τ, so τ is compared with a published relation below.

Table 4: Validation at $1.4\,M_\odot$ against six published piecewise-polytropic EoS fits. R and Λ were used to calibrate one crust parameter; f was held out.

EoS	R (km)	$R_{\rm ref}$	Λ	$\Lambda_{\rm ref}$	f (Hz)	$f_{\rm ref}$	$\Delta f/f$ (%)	τ (s)
SLy	11.7055	11.7061	295.5	295.6	1935.20	1935.06	+0.0072	0.1920
AP3	11.9712	11.9717	367.1	367.2	1848.51	1848.40	+0.0059	0.2094
AP4	11.3192	11.3197	247.5	247.6	2007.63	2007.50	+0.0062	0.1793
ENG	11.9695	11.9699	369.9	370.1	1845.65	1845.53	+0.0066	0.2100
MPA1	12.4339	12.4343	478.9	479.1	1745.68	1745.58	+0.0057	0.2341
WFF2	11.1245	11.1249	224.9	225.0	2047.47	2047.34	+0.0062	0.1729

Analytic benchmarks with a finite surface density. Three exact Newtonian results for the homogeneous incompressible star test the solver with a density jump at the surface, as in self-bound strange matter: the Kelvin f-mode, $\omega^2 = 2l(l-1)/(2l+1)\,GM/R^3 = 0.8\,GM/R^3$ for $l = 2$, the tidal Love number $k_2 = 3/4$, and the quadrupole-formula damping time of the Kelvin mode, $\tau_N = 19.53125\,R^4/M^3$ ($G = c = 1$). The GR solver was run on a nearly incompressible constant-density EoS ($c_s^2 = 10^4$) at compactnesses $C = M/R$ from 0.002 to 0.04; the deviations from the Newtonian values vanish linearly with C as they must (at $C = 0.002$ the frequency exceeds the Kelvin value by 2.3e-04, and the linear extrapolation to $C \to 0$ gives 1e-04). For the Love number, the matching at the surface requires the correction for the discontinuity in the density (Postnikov et al., 2010; Damour & Nagar, 2009): with it $k_2 = 0.742$ at $C = 0.002$ and the extrapolation to $C \to 0$ is within 0.2% of 3/4 (slope -5.3 per unit C), whereas without it $k_2 = 0.0004$. The damping time approaches τ_N in the same way (the deviation extrapolates to 1e-04 at $C \to 0$, slope 4.7 per unit C), which tests the interior–exterior coupling that produces $\mathrm{Im}\,\omega$ in the weak-field limit. This tests the f-mode, damping-time and tidal matching at a finite surface density, but only in the Newtonian limit; the numerical convergence for self-bound stars (the MIT bag and a pQCD-form EoS) gives changes of at most 3e-05 in f and 4e-04 in τ for the same variations of the numerical settings as above.

Comparison with a published full-GR relation for τ. Lioutas & Stergioulas (2018) fitted the full-GR damping times of nine nuclear EoS for $M/R > 0.1$ with $M/\tau = 0.112x^4 - 0.53x^5 + 0.628x^6$ ($x = M/R$, $G = c = 1$; their Eq. 20, quoted correlation coefficient 0.9997). At $1.4\,M_\odot$ the damping times of the six reference EoS (compactness 0.166–0.186) lie **2.9 to 7.0% above** this relation, and the relation of Andersson and Kokkotas (their Eq. 18 as quoted by Lioutas & Stergioulas 2018) is exceeded by 6–9%. In the nuclear populations the median offsets are +4% (nucleonic), +5% (dark-matter) and +1% (hyperonic-like) for population A and +11 to +12% for population B; self-bound stars lie far outside the relation (+51% and +49%), which is expected because it was fitted to nuclear EoS without a surface density jump. The origin of the 3–7% offset for the reference EoS has not been identified; the Newtonian-limit benchmark above and the exterior black-hole benchmark rule out a gross error in the damping-time machinery but not a few-per-cent error at stellar compactness (possible causes include the crust model, EoS differences from the fitting sample and the accuracy of the fit itself, none of which was tested). The absolute damping times should therefore be regarded as uncertain at the level of several per cent; because τ adds no classification information in any experiment, no conclusion of this paper depends on it.

3.3 The data set

Table 5 and Fig. 4 summarise the 360 EoS (2886 computed modes in total; about 199 s of computation per EoS). The strange family lies on a distinct f–R relation, whereas the nucleonic, dark-matter and hyperonic-like families overlap heavily; the hyperonic-like family extends to higher frequencies.

Table 5: Population data set (90 EoS per scenario after screening): ranges of $M_{\max}$ ($M_\odot$), $R_{1.4}$ (km), $\Lambda_{1.4}$ and f at $1.4\,M_\odot$ (Hz).

Scenario	N	$M_{\max}$	$R_{1.4}$	$\Lambda_{1.4}$	$f_{1.4}$
Nucleonic	90	1.97–2.59	10.9–13.8	218–682	1614–2061
Strange	90	2.02–3.34	10.0–11.7	265–797	1558–1979
Dark-matter	90	1.97–2.66	10.8–13.9	199–735	1587–2098
Hyperonic	90	1.97–2.30	10.1–13.5	104–663	1625–2380

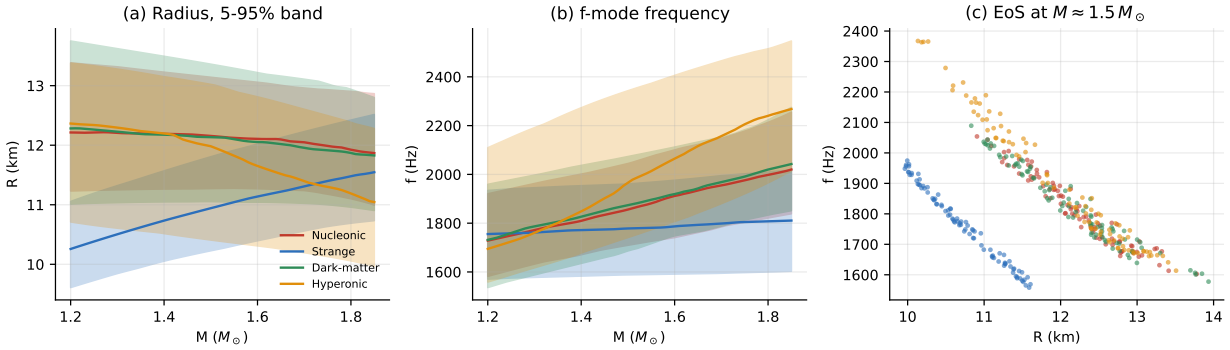


Figure 4: Population data set. (a) Radius and (b) f-mode frequency versus mass; lines are medians and bands the 5–95% range across EoS of each scenario. (c) f versus R at $1.5\,M_\odot$, one point per EoS.

4 Methods

4.1 Whole-EoS validation

Every EoS belongs to exactly one of five folds (stratified by scenario). In each of five rounds one fold is the test set, the next fold is the calibration set and the other three are the training set; no EoS appears in more than one role, so every reported number is for EoS the classifier has never seen. The partition is repeated for three random seeds; we report the mean over folds within a repetition and Student-t 95% intervals over the three repetitions. Observations are (M, R, f, τ) at masses drawn uniformly over 1.2–1.85 $M_\odot$ (a mass prior shared by all classes), with independent noise draws, and classes are balanced (chance 25%).

4.2 Observational-error scenarios

Observables are perturbed multiplicatively, $x_j^{\rm obs} = x_j(1 + \sigma_j \epsilon_j)$, $\epsilon_j \sim \mathcal{N}(0,1)$ independent. Draws with a non-positive factor (only possible for the damping time at large s: 0.6% of the draws at $s = 2$ and 11% at $s = 4$; none elsewhere) are redrawn, so the noise is a truncated Gaussian; the kernel-Bayes likelihood ignores this truncation. *Literature-anchored scenarios:* **NICER-like**: $\sigma_M = 10\%$, $\sigma_R = 9\%$, no f or τ; these are the 68% uncertainties of the NICER analyses of PSR J0030+0451 ($M = 1.34^{+0.15}_{-0.16}$ and $1.44^{+0.15}_{-0.14}\,M_\odot$, $R = 12.71^{+1.14}_{-1.19}$ and $13.02^{+1.24}_{-1.06}$ km; Riley et al. 2019; Miller et al. 2019). **NICER-like plus f-mode**: in addition $\sigma_f = 20\%$, motivated by the forecast that third-generation detectors could measure the f-mode frequency of a GW170817-like inspiral to within a few hundred Hz (Williams et al., 2022), i.e. roughly 10–25% of f. These two are illustrative combinations drawn from different sources (a NICER pulsar and a hypothetical GW170817-like inspiral), not a realistic joint observation: mass, radius and f-mode of a single object are not currently available together. *Precision scale:* to show what would be required, the same observables are also evaluated with errors $s \times (5\%, 5\%, 2\%, 20\%)$ in (M, R, f, τ) for $s \in \{0.25, 0.5, 1, 2, 4\}$, and in the noise-free limit. These values are not forecasts for any instrument; they define a scale on which the literature-anchored scenarios can be placed. Three observable sets are used: $\{M, R\}$, $\{M, R, f\}$ and $\{M, R, f, \tau\}$.

4.3 Classifiers

Learners: logistic regression, k-nearest neighbours ($k = 25$), a random forest (200 trees, minimum leaf 5), histogram gradient boosting, an RBF support-vector machine (8000 training samples) and an MLP deep ensemble (Lakshminarayanan et al., 2017) of three 2×64 ReLU networks (the architecture of the earlier version) on log-transformed, standardised inputs. Training data contain three noise realisations of 33 masses for each training EoS.

4.4 Kernel-Bayes reference classifier

For a reference set of (EoS, M) points the posterior of class k for an observation $\mathbf{x}^{\rm obs}$ is

$$P(k \mid \mathbf{x}^{\rm obs}) \;\propto\; \frac{1}{N_k} \sum_{i \in k} \prod_{j} \mathcal{N}\Big(x_j^{\rm obs};\, x_{ij},\, (\sigma_j x_{ij})^2\Big)\,, \tag{1}$$

which is the Bayes-optimal classifier for the stated error model and a uniform prior over the reference EoS and masses. We use the training EoS as reference (the same information as the learners) and, separately, all EoS except the test fold (a larger reference). Unlike the single-sequence oracle of Appendix A, this is not an upper bound, because the test EoS are not in the reference set; agreement between learners and kernel-Bayes indicates that the remaining error is set by the information in the observables rather than by the learner.

4.5 Uncertainty quantification

Calibration: expected calibration error (ECE, 15 bins), negative log-likelihood (NLL) and Brier score, before and after temperature scaling fitted on the calibration EoS. *Epistemic uncertainty*: mutual information across ensemble members. *Conformal prediction*: split conformal prediction with the score $1 - \hat{p}(y \mid x)$ at target miscoverage $\alpha = 0.1$ (Vovk et al., 2005; Angelopoulos & Bates, 2023), calibrated on the calibration EoS with *one star per calibration EoS*: for each of the $n \approx 72$ calibration EoS one star (random mass, one noise draw) is chosen, so that the calibration units are independent (EoS, star) pairs that are exchangeable with a test star drawn the same way from a new EoS of the same population. The score quantile is taken at level $\lceil (n+1)(1-\alpha) \rceil / n$, as in standard split conformal prediction, and the reported sets and coverage are averages over 40 random choices of the calibration stars. The textbook finite-sample guarantee requires the calibration and test units to be exchangeable. Here the folds are stratified and class-balanced: the calibration fold contains a fixed number of EoS per class, the test EoS come from a disjoint fold with the same composition, and both are drawn without replacement from a finite catalogue. Exchangeability then holds within each class and for the class-balanced mixture only approximately (conditioning on the fixed class counts of the calibration fold breaks it by an amount of order $1/n$), so we do not claim the exact distribution-free guarantee; the coverage reported is an *empirical marginal coverage* for stars of unseen EoS drawn from the same catalogue distribution with a uniform class prior. As a sensitivity check, a pooled variant that uses all stars of the calibration EoS (33 masses, two noise draws each) with the same n is also reported; its stars are correlated, so for it even the approximate exchangeability argument fails. The coverage is marginal, not conditional on class or mass, and assumes that the EoS of the catalogue are exchangeable draws from one population; it does not extend to EoS from another population (Sec. 5.8).

5 Results

5.1 Noise-free separability of EoS populations

Figure 5 shows the held-out-EoS accuracy without measurement error, and the left panel of Fig. 7 the confusion matrix. With M and R only the accuracy is 39% (MLP ensemble; logistic regression 36%, random forest 39%); adding the f-mode raises it to 65% and adding τ changes nothing

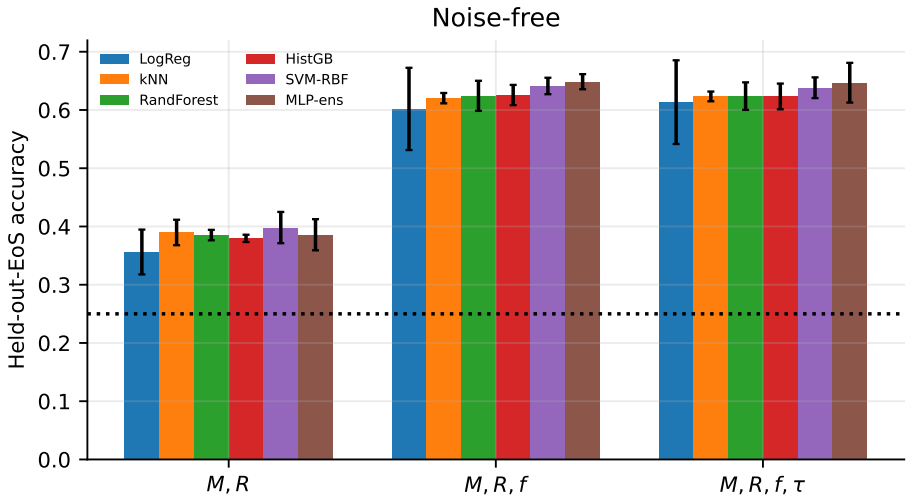


Figure 5: Held-out-EoS accuracy without measurement error for each learner and observable set (95% t-intervals over three repetitions; dotted: chance).

Table 6: Held-out-EoS accuracy (%, mean ± 95% t-interval over three repetitions) by error scenario and learner. Chance is 25%.

Scenario	LogReg	k-NN	RF	HistGB	SVM	MLP ens.	Kernel Bayes
NICER-like: M, R at 10%, 9%	33±1	32±0	31±0	33±1	34±0	33±1	34±1
+ f-mode at 20%	36±1	35±1	36±1	37±1	37±1	37±1	37±1
$s = 1$: 5%, 5%, 2%	50±3	52±1	52±1	52±2	52±4	53±3	53±2
$s = 1$ with τ 20%	50±2	52±1	52±2	52±3	52±3	53±3	53±2
$s = 0.25$ (1.3%, 1.3%, 0.5%, 5%)	58±5	60±2	60±2	60±2	60±4	61±2	60±1

(65%). All learners agree within a few points. The confusion matrix shows why the ceiling is so low: strange matter is recognised perfectly (100%), hyperonic-like matter in 76% of cases, but nucleonic and dark-matter-admixed stars are assigned to each other almost at random (recall 50% and 33%; accuracy between the two classes 51%, chance 50%). Merging them into one class gives 84% three-class accuracy; the baseline for this merged problem is 50% (always predicting the merged class, which holds half of the stars), not the 33% of three balanced classes. The reason is physical: the dark component shifts f by only 0.57% at fixed nuclear EoS (median over the population; 25.7 Hz or 1.3% for SLy with $G = 100\,\mathrm{fm}^2$; Sec. 5.5), far less than the spread of f at fixed mass produced by the uncertain nuclear EoS (about ±10%; Table 5).

5.2 Errors anchored on published measurements

Table 6 gives the accuracy under the literature-anchored scenarios and at selected points of the precision scale. With NICER-like errors (M 10%, R 9%) the accuracy is 33% (MLP ensemble; kernel-Bayes 34%), barely above the 25% chance level; adding a 20% f-mode measurement raises it to 37%. The three-class accuracy after merging nucleonic and dark-matter-admixed matter is 51%, no better than the 50% majority-class baseline of the merged problem. Errors of 5%, 5% and 2% in (M, R, f) would give 53%, and 1.3%, 1.3%, 0.5% would give 61%, which approaches but never exceeds the noise-free ceiling of 65%.

5.3 Precision scale and which observable matters

Figure 6 shows the accuracy against the precision scale. For the MLP ensemble, $\{M, R\}$ gives 38%, 37% and 30% at $s = 0.25$, 1 and 4, whereas $\{M, R, f\}$ gives 61%, 53% and 38%; the damping time with a 20% error (scaled with s) adds nothing within the uncertainty. Learners and kernel-Bayes

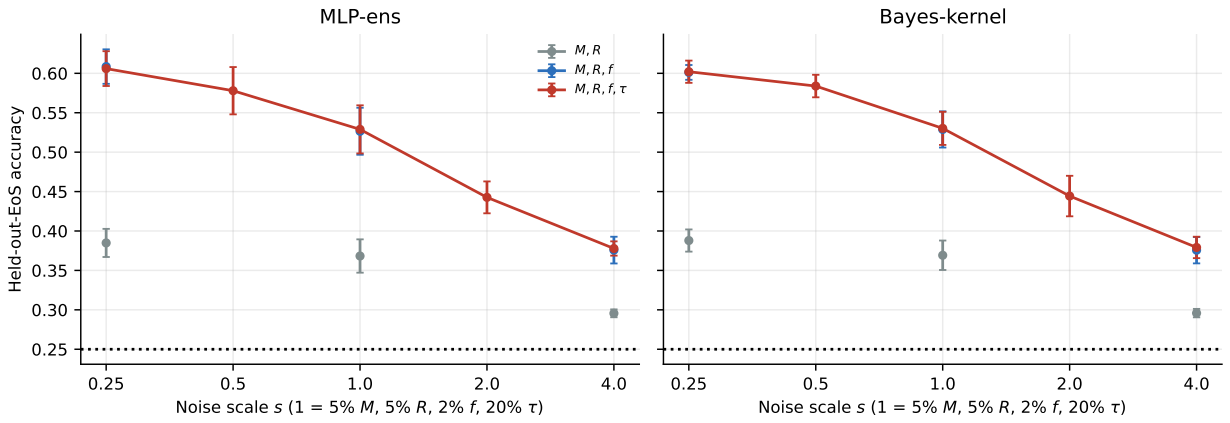


Figure 6: Held-out-EoS accuracy versus precision scale s ($1 = 5\%, 5\%, 2\%, 20\%$ in M, R, f, τ) for the MLP ensemble and the kernel-Bayes classifier, for three observable sets (95% t-intervals; dotted: chance). The $\{M, R, f\}$ and $\{M, R, f, \tau\}$ curves coincide.

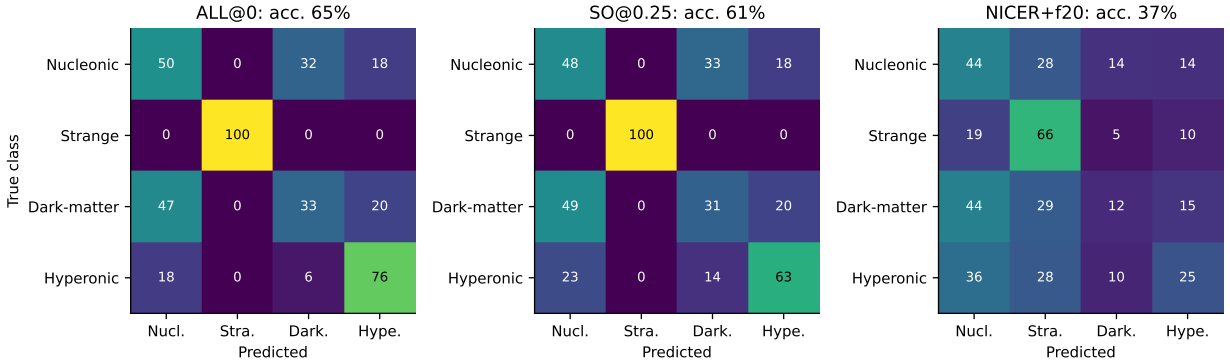


Figure 7: Row-normalised confusion matrices (%) of the temperature-scaled MLP ensemble, pooled over folds and repetitions. Left to right: noise-free, $s = 0.25$, and NICER-like errors plus a 20% f-mode measurement.

agree to a few points at every scale (at $s = 1$: MLP 53%, kernel-Bayes 53%, kernel-Bayes with all other EoS as reference 53%; logistic regression 50%, random forest 52%, gradient boosting 52%, SVM 52%), so none of the learners tried leaves a measurable amount of information unused, and enlarging the reference sample from the training EoS to all other EoS does not change the result (this is evidence for an information plateau within these priors, not a proof of an upper bound).

5.4 Degeneracies

Figure 7 shows row-normalised confusion matrices. With accurate f ($s = 0.25$) strange matter is still identified perfectly (100%) and the hyperonic-like class in 63%, but the nucleonic and dark-matter rows remain mixed (48% and 31% correct, 33% and 49% assigned to each other). With a 20% f-mode error all rows are confused with all columns, strange matter being the best recovered (66%).

5.5 Dark matter when the nuclear EoS is known

To separate "the dark component is small" from "the nuclear EoS is unknown", the observables of each of the 90 dark-matter-admixed EoS were also computed for its own nuclear base EoS without the dark component, at the same masses. At fixed nuclear EoS and mass the dark component raises f by a median of 0.57% (EoS-level bootstrap interval of the median 0.50–0.66%, conditional on the construction below; 5–95% range 0.27–2.22%; positive for 100% of the pairs) and lowers R

by a median of -0.40% (-1.50 to -0.20%); the shift falls steeply with the self-interaction (Fig. 8a; correlation of $\ln G$ with $\Delta f/f$ at $1.4\,M_\odot$ is -0.95). *The shift is not independent of the model construction:* when the baryon-number normalisation of the one-fluid barotrope is changed to its extreme value, the shift of an individual EoS changes by a median of 5% and at most 6% (and by up to 65% for EoS with m_χ above the stability window in the wide-prior sample; see the paragraph on the dark-matter construction below), and the bootstrap interval reflects the EoS sample, not this modelling uncertainty. If the observer were told the nuclear EoS exactly, the Bayes accuracy for dark matter versus no dark matter (equal priors, independent Gaussian errors, mass identical by construction) would reach 75% at $\sigma_f = 0.51\%$ (EoS-level bootstrap interval 0.45–0.59%) when R is known to 1% (0.51% for 5%) and 90% at $\sigma_f = 0.20\%$ (0.17–0.23%) (Fig. 8b); at the points of the precision scale it is 76% at $s = 0.25$ (0.5% in f, 1.25% in R), 58% at $s = 1$ and 52% at $s = 4$, and with a 20% f-mode error and 9% in R it is 52%. With the nuclear EoS *unknown* the accuracy between the nucleonic and dark-matter classes is 51% even without any measurement error (Sec. 5). The lack of discrimination therefore comes from the uncertainty of the nuclear EoS, not from the smallness of the shift alone: detecting the dark component needs both an f-mode measurement at the 0.5% level and a nuclear EoS known to better than the shift itself.

Sensitivity to the dark-matter construction. The one-fluid barotrope assigns baryon number through $n_B = \rho/m_0$. This paper uses $m_0 = 931.494\,\mathrm{MeV}$ (the atomic mass unit, the normalisation of the polytropic fits); the companion pilot (Husain, 2026b) uses $m_0 = 931.192\,\mathrm{MeV}$. The baryon chemical potential $\mu_N \to m_0$ at low density, so m_0 fixes where μ_N exceeds m_χ and the dark component appears, and it is not a free parameter of the dark-matter physics but a convention of the effective barotrope. The two values give the same shift to within 0.2% for a typical EoS (at most 0.2% over the 90 EoS tested; for SLy with $G = 100\,\mathrm{fm}^2$, 25.7 Hz here against 25.8 Hz in the pilot, a difference that also includes the different solvers), so the reconciliation of the two normalisations does not matter at the level of the headline number. Larger changes of convention move the population median very little: with $m_0 = 935$ and $938.9\,\mathrm{MeV}$ (the latter places the threshold at zero density) the median shifts at $1.4\,M_\odot$ are 0.51% and 0.51% against 0.50% (5–95% ranges 0.27–2.04 and 0.28–2.09% against 0.26–1.99%), but for an individual EoS of the primary (in-window) population the shift changes by a median of 5% and at most 6% between the default and the extreme prescription. The sensitivity is much larger when m_χ lies above the stability window: for the 7 EoS of the earlier wide-prior sample with $m_\chi > 938.783\,\mathrm{MeV}$ (of 16 tested) the change has a median of 9% and a maximum of 65%, the largest changes (23–65%) occurring for three EoS with $m_\chi = 939.00$–$939.03\,\mathrm{MeV}$, just above the extreme normalisation of $938.9\,\mathrm{MeV}$; the in-window EoS of that sample change by at most 6%. The shift depends much more on G (correlation of $\ln G$ with $\Delta f/f$: -0.95) than on m_χ (correlation -0.04). The qualitative conclusion is unchanged, but the shift of a single EoS depends on the prescription at the per-cent level inside the window (and strongly outside it), so the quantitative detectability thresholds above (e.g. the f-mode precision for 75% accuracy) carry this modelling uncertainty in addition to their statistical intervals, and the effective one-fluid oscillation model (equilibrium composition, common displacement of the two components) remains an explicit assumption of every dark-matter statement in this paper.

5.6 Calibration and prediction sets

The probabilities are well calibrated because the learners are close to the information limit (Fig. 9a): at $s = 1$ the ECE is 0.041 for the raw ensemble, 0.040 after temperature scaling and 0.041 for kernel-Bayes, with NLL 0.92 nats for all three (Table 7). Conformal sets calibrated with one star per EoS have empirical marginal coverage of 0.91–0.92 at the 0.90 target for every scenario (as expected for the finite-sample-corrected quantile; no formal guarantee is claimed for the stratified folds, Sec. 4.5); the pooled variant gives 0.91–0.92 with almost identical sets (2.24 against 2.25 at $s = 1$). The price of validity is set size: 3.32 of four classes under NICER-like errors, 3.22 when a 20% f-mode is added, 2.25 at $s = 1$ and 1.87 at $s = 0.25$. Temperature scaling and kernel-Bayes

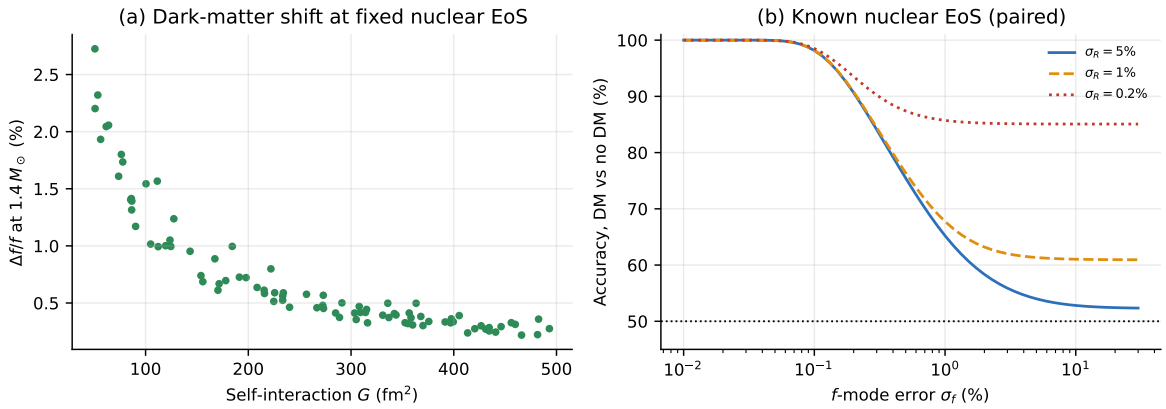


Figure 8: Dark matter at fixed nuclear EoS (90 paired EoS). (a) Fractional f-mode shift at $1.4\,M_\odot$ versus the self-interaction G. (b) Bayes accuracy for dark matter versus no dark matter, averaged over EoS and masses, as a function of the f-mode error for three radius errors (dotted: chance).

Table 7: Calibration and conformal sets (mean over three repetitions): ECE and NLL (nats) for the raw MLP ensemble, the temperature-scaled ensemble ($+T$) and kernel-Bayes, and coverage and set size (target coverage 0.90) of the conformal sets calibrated with one star per EoS.

	ECE			NLL			MLP $+T$		Kernel Bayes	
Scenario	Raw	$+T$	KB	Raw	$+T$	KB	Cov.	Size	Cov.	Size
NICER-like	0.022	0.025	0.021	1.32	1.32	1.32	0.92	3.32	0.92	3.31
$+$ f at 20%	0.021	0.021	0.021	1.29	1.29	1.29	0.92	3.22	0.92	3.23
$s = 0.25$	0.057	0.045	0.099	0.71	0.71	0.82	0.92	1.87	0.92	1.94
$s = 0.5$	0.053	0.045	0.060	0.75	0.75	0.77	0.92	1.96	0.92	1.99
$s = 1$	0.041	0.040	0.041	0.92	0.92	0.92	0.91	2.25	0.92	2.27
$s = 2$	0.035	0.040	0.031	1.16	1.16	1.15	0.92	2.88	0.92	2.87
$s = 4$	0.025	0.029	0.023	1.29	1.29	1.29	0.92	3.27	0.92	3.25

give the same sets. At realistic precision the data therefore support a short list containing most of the classes, not a single label.

5.7 Statistical robustness

EoS-level bootstrap. The intervals of the preceding sections describe re-partitioning of the same 360 EoS. Table 8 instead gives 95% intervals from resampling whole EoS within each class (2000 resamples; each EoS's predictions pooled over the three partitions). These intervals are *conditional* on the catalogue and the trained classifiers: they quantify which EoS happen to be in the test sets, but not the variability from regenerating the EoS populations from the priors, from retraining the classifiers on a different catalogue, or from changing the priors themselves (the last two are only partly probed by the learning curves and by population B). The noise-free accuracy of the MLP ensemble is 64% [61, 67], the recall of strange matter is 100% [100, 100], and the accuracy between the nucleonic and dark-matter classes is 51% [45, 56]; with NICER-like errors and a 20% f-mode it is 50% [47, 52] and the overall accuracy 37% [36, 38]. The intervals for the nucleonic–dark-matter pair contain 50% in every scenario, so the absence of demonstrated discrimination is not an artefact of a particular partition of this finite sample; it remains conditional on the catalogue and priors, and an interval containing 50% does not prove that the classes are identical, only that these experiments do not show a difference.

Population size. Figure 10 shows the accuracy as a function of the number of training EoS (54 to 216). Without noise the logistic regression is already at 62% with 54 training EoS and the MLP

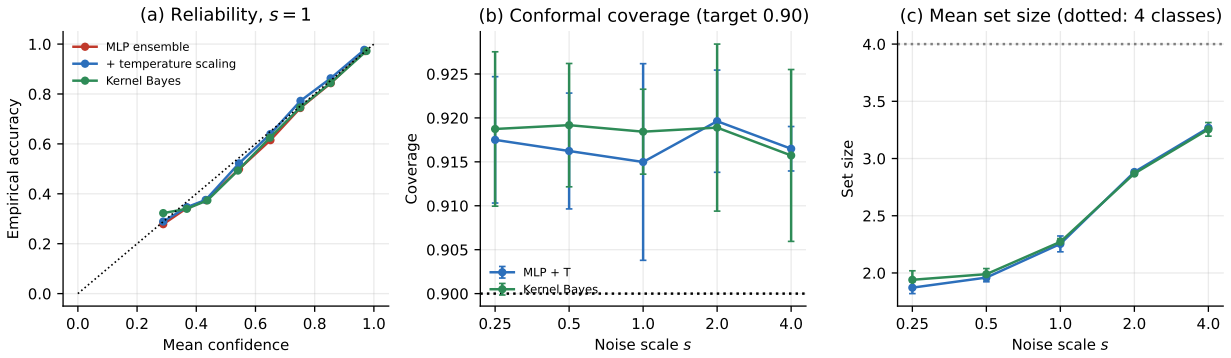


Figure 9: (a) Reliability diagram ($s = 1$, pooled). (b) Conformal coverage and (c) mean prediction-set size versus precision scale (95% t-intervals).

Table 8: EoS-level bootstrap estimates (%, 95% interval): overall accuracy, accuracy between the nucleonic and dark-matter classes (chance 50%) and recall of strange matter.

Scenario	Learner	Accuracy	Nucleonic vs DM	Strange recall
ALL 0	MLP-ens	64 [61, 67]	51 [45, 56]	100 [100, 100]
ALL 0	LogReg	61 [58, 64]	50 [45, 54]	100 [100, 100]
ALL 0	RandForest	62 [60, 65]	52 [48, 57]	100 [100, 100]
NICER+f20	MLP-ens	37 [36, 38]	50 [47, 52]	66 [64, 67]
NICER+f20	LogReg	36 [34, 37]	49 [46, 53]	61 [59, 62]
NICER+f20	RandForest	36 [35, 37]	50 [50, 51]	59 [57, 60]
NICER+f20	Bayes-kernel	37 [36, 38]	49 [47, 52]	67 [65, 68]
SO 1.0	MLP-ens	52 [50, 55]	48 [45, 51]	90 [89, 92]
SO 1.0	LogReg	50 [47, 53]	48 [43, 52]	90 [88, 91]
SO 1.0	RandForest	52 [51, 54]	50 [48, 51]	89 [88, 91]
SO 1.0	Bayes-kernel	53 [51, 55]	49 [46, 51]	90 [89, 92]
SO 0.25	MLP-ens	61 [58, 63]	49 [45, 54]	100 [100, 100]
SO 0.25	LogReg	58 [55, 60]	48 [44, 53]	100 [100, 100]
SO 0.25	RandForest	60 [58, 62]	50 [46, 53]	100 [100, 100]
SO 0.25	Bayes-kernel	60 [58, 62]	50 [47, 53]	100 [100, 100]

ensemble at 62% with 108, against 64% with 216; under NICER-like errors plus a 20% f-mode the accuracy is 35–37% at every size. The plateau therefore does not appear to result from too few EoS, and a larger population of the same kind is unlikely to change the numbers.

Wider dark-fermion mass prior (sensitivity). The primary population restricts m_χ to the stability window $937.9 \leq m_\chi \leq 938.783\,\mathrm{MeV}$. As a sensitivity experiment the earlier population, whose dark-matter EoS drew m_χ up to 939.2 MeV (27 of its 90 dark-matter EoS above the upper bound and 63 inside), was analysed with the same protocol; the other three classes are identical. The results are essentially the same as for the primary population: 64% without noise, 61% at $s = 0.25$, 53% at $s = 1$ and 37% for NICER-like errors plus a 20% f-mode (MLP ensemble), with an accuracy of 52% between the nucleonic and dark-matter classes (primary: 65%, 61%, 53%, 37% and 51%).

5.8 Robustness to the EoS populations

The preceding results depend on the parametrisations and priors of Table 3. To test this, an alternative population ("B") was constructed with different functional forms and priors (Table 9); the same screens, solver and experiments were applied (48 EoS per scenario, 192 in total; Table 10). Population B is still parametric rather than microphysical, so it tests sensitivity to the parametrisation, not to the underlying nuclear physics.

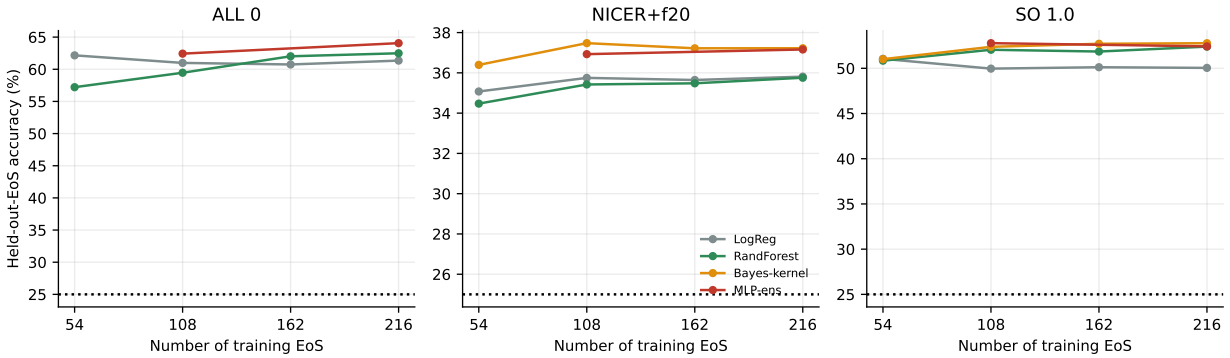


Figure 10: Held-out-EoS accuracy versus the number of training EoS for four learners (pooled over repetitions; dotted: chance). The MLP ensemble is shown at 108 and 216 training EoS only.

Table 9: Alternative population B. ε_x is the energy density at $1.3\,n_{\rm sat}$ ($\rho_x = 10^{14.55}\,\mathrm{g\,cm^{-3}}$); sound-speed profiles are piecewise linear in $\ln\varepsilon$ through knots at $\varepsilon/\varepsilon_x = 1, 1.5, 2.5, 4, 7$ and constant above; the lower-density EoS is a single polytrope on the same crust.

Scenario	Parameters (uniform priors)
Nucleonic	polytrope $\log_{10} P_1 \in [34.15, 34.55]$, $\Gamma_1 \in [2.6, 3.6]$ up to ρ_x; then $c_s^2 \in [0.1, 1]$ at each knot (first knot $[0.1, 0.6]$)
Hyperonic-like	as nucleonic, but $c_s^2 \in [0.15, 0.6]$ at the first two knots, a dip $c_s^2 \in [0.03, 0.3]$ at $2.5\,\varepsilon_x$ and $c_s^2 \in [0.5, 1]$ at the last two
Strange (self-bound)	$P(\mu) = 3a_4\mu^4/(4\pi^2) - 3a_2\mu^2/(4\pi^2) - B$ with $a_4 \in [0.55, 1]$, $\sqrt{a_2} \in [0, 150]\,\mathrm{MeV}$, $B^{1/4} \in [135, 165]\,\mathrm{MeV}$
Dark-matter admixed	nucleonic B plus a dark Fermi gas with $m_\chi \in [937.9, 938.78]\,\mathrm{MeV}$ (inside the stability window) and $G \in [50, 500]\,\mathrm{fm}^2$

Within population B. With whole-EoS hold-out, the MLP ensemble reaches 49% without noise (logistic regression 52%, random forest 52%), against 65% for population A. Strange matter is again recognised perfectly (100%) and the accuracy between the nucleonic and dark-matter classes is again at the level of chance (51%); nucleonic matter is recognised in 34%, dark-matter in 37% and hyperonic-like matter in only 26% (76% in population A), so the separability of the hyperonic-like family, and with it the overall plateau, depends on how the family is built. At $s = 0.25$ and $s = 1$ the accuracies are 50% and 45% (A: 61% and 53%), and for NICER-like errors plus a 20% f-mode 37% (A: 37%); there strange matter is recognised in 74% of cases.

Out-of-family transfer. Table 12 gives the accuracy of classifiers trained on all EoS of one population and tested on all EoS of the other. Without noise the transfer accuracy is 54% (A→B) and 54% (B→A) for the MLP ensemble, between the within-population values, and with NICER-like errors plus a 20% f-mode 36% and 33%. Strange matter is still recognised in 100% and 100% of cases without noise, and the accuracy between the nucleonic and dark-matter classes remains 52% and 53%. The results that survive a change of population are therefore (i) the absence of demonstrated nucleonic–dark-matter discrimination, (ii) an accuracy of about 33–37% under NICER-like errors, with or without a 20% f-mode, and (iii) the identifiability of self-bound strange matter once f is accurate; the 65% plateau and the identifiability of the hyperonic-like family are specific to the construction of population A.

Prior width. To probe how the nucleonic–dark-matter accuracy depends on the width of the nuclear prior, the main population was restricted to EoS whose four polytropic parameters all lie in the central fraction w of their prior ranges (the same restriction for both classes; Table 11). For $w = 1$, 0.9, 0.8 and 0.7 (down to 32 nucleonic and 23 dark-matter EoS) the accuracy of logistic regression and of a random forest stays between 45% and 61% in the noise-free case and at $s = 0.25$

Table 10: Population B (48 EoS per scenario): ranges of $M_{\max}$ ($M_\odot$), $R_{1.4}$ (km), $\Lambda_{1.4}$ and f at $1.4\,M_\odot$ (Hz).

Scenario	N	$M_{\max}$	$R_{1.4}$	$\Lambda_{1.4}$	$f_{1.4}$
Nucleonic	48	2.04–3.05	11.5–13.5	309–795	1563–1913
Strange	48	1.97–2.20	10.9–11.7	441–761	1575–1776
Dark-matter	48	2.01–3.05	11.6–13.7	287–798	1557–1947
Hyperonic	48	2.02–2.51	11.9–13.5	290–797	1556–1942

Table 11: Accuracy (%) between the nucleonic and dark-matter classes versus the central fraction w of the nuclear parameter ranges (10 repetitions of 5-fold whole-EoS cross-validation; chance 50%). N is the number of nucleonic + dark-matter EoS.

Errors	w	N	LogReg	RF
noise-free	1	90+90	45	51
noise-free	0.9	58+59	45	51
noise-free	0.8	47+36	54	54
noise-free	0.7	32+23	51	61
s =0.25	1	90+90	45	49
s =0.25	0.9	58+59	47	49
s =0.25	0.8	47+36	53	51
s =0.25	0.7	32+23	56	53
s =1	1	90+90	46	49
s =1	0.9	58+59	46	48
s =1	0.8	47+36	55	51
s =1	0.7	32+23	54	50

and $s = 1$. The highest value (61%, random forest, noise-free, $w = 0.7$) rests on 55 EoS, for which the binomial standard error is about 7 points, and it is not accompanied by a rise in the other settings or for logistic regression, so the values are compatible with chance at these sample sizes; narrowing the prior by 30% per parameter does not make the dark component recognisable in these experiments. Substantially narrower priors would leave too few EoS in this catalogue, and the limit of a known nuclear EoS ($w \to 0$), where the dark component can be detected at the sub-per-cent level, is treated in Sec. 5.5.

6 Discussion and limitations

What the results support. (i) A high accuracy on single-sequence data does not carry over to populations: with whole-EoS hold-out the highest accuracy reached by any method for these four scenarios is about 65% even without measurement error (49% in the alternative population B). (ii) The plateau looks informational: six different learners (including the MLP ensemble) and a kernel-Bayes classifier built on the exact error model agree, so within these priors a better learner is unlikely to help (the kernel-Bayes classifier uses a finite reference population and is not a rigorous upper bound). (iii) With the errors of current X-ray measurements and a plausible third-generation f-mode measurement, composition is essentially not identified in these experiments (33–37% against 25% chance); an accurate f-mode is the observable that would help, and the damping time adds nothing. (iv) In the populations tested, no discrimination between dark-matter-admixed and nucleonic matter is demonstrated unless the nuclear EoS is known to better than the shift in f that the dark component produces (median 0.57%, up to about 2% for weak self-interaction); with the nuclear EoS known exactly, an f-mode precision of about 0.51% would give 75% accuracy (Sec. 5.5), in line with the EoS-conditional analysis of the earlier pilot. (v) Probabilities and conformal sets are reliable and turn the lack of discrimination into an explicit statement about which classes remain plausible.

Table 12: Out-of-family test: accuracy (%, mean ± 95% t-interval over three noise realisations) of classifiers trained on all EoS of one population and tested on all EoS of the other. Chance is 25%.

Train→test	Scenario	LogReg	RF	HistGB	MLP ens.	Kernel Bayes
A→B	ALL 0	54±0	50±0	50±4	54±1	–
A→B	NICER+f20	35±1	34±1	36±1	36±1	36±0
A→B	SO 1.0	42±0	39±1	39±1	40±1	40±1
A→B	SO 0.25	52±0	48±1	50±4	53±2	48±0
B→A	ALL 0	51±0	51±2	52±0	54±1	–
B→A	NICER+f20	32±1	34±1	34±1	33±1	35±1
B→A	SO 1.0	41±0	46±0	46±1	48±0	46±0
B→A	SO 0.25	50±0	51±1	51±2	54±1	52±0

Limitations. *Phenomenological families.* The EoS families are parametrisations with uniform priors, screened by $M_{\max}$, causality, $\Lambda_{1.4}$ and $R_{1.4}$; conclusions depend on these priors. The absence of demonstrated nucleonic–dark-matter discrimination in particular follows from the dark-matter scenario sharing the nuclear prior. Microphysical models (for example a population of QMC EoS) would change the priors and might make some classes easier or harder to separate. The hyperonic-like family is a soft-then-stiff parametrisation, not a hyperonic calculation, and the strange family uses a constant speed of sound; the classifier therefore separates parametrisation families. A second population with different functional forms and priors (Sec. 5.8) again shows no demonstrated nucleonic–dark-matter discrimination, a similar NICER-like accuracy and the recognition of strange matter (self-bound models with a surface density jump give a distinct f–R relation in both), but not the 65% plateau or the identifiability of the hyperonic-like family. No test with microphysically computed EoS (for example hyperonic or quark-matter calculations) has been made, and the conclusions about composition remain conditional on parametric families. *Dark matter.* The dark component is treated as an effective one-fluid barotrope in chemical equilibrium; the check against the two-fluid pilot is a single number (25.7 versus 25.8 Hz), a frozen-composition response changes the shift by only 1.1% (Sec. 3.2), but the assignment of baryon number to the fitted barotrope changes the shift of an individual EoS by up to 6% inside the stability window and by up to 65% for dark-fermion masses above it (Sec. 5.5). Independent displacement of the two fluids (a genuine two-fluid eigenmode calculation), other dark-sector models and other dark fractions are not explored. *Solver.* Non-rotating stars, $l = 2$ only, adiabatic perturbations equal to the equilibrium barotrope (no composition gradients, entrainment or crust elasticity); the damping time is checked only through the exterior black-hole benchmark, the Newtonian-limit benchmark and a published universal relation, from which it differs by 2.9–7.0% for the reference EoS (unexplained); the crust has one calibrated parameter; and no comparison with published numerical f-mode tables for realistic EoS has been made (the six reference values come from an unpublished same-author pilot). *Errors.* Independent multiplicative Gaussian errors are used; the NICER and f-mode values are anchored on published results and forecasts for individual sources, not on a population of detections, and systematic theory error and correlated errors are not modelled. *Statistics and sampling.* 90 EoS per scenario (48 in population B) and three fold partitions that reuse the same EoS, so the t-intervals reflect partition and noise variability only; EoS-level bootstrap intervals and learning curves are given in Sec. 5.7. A uniform mass prior over 1.2–1.85 $M_\odot$ is used; the conformal coverage is empirical and marginal only (no exact guarantee is claimed for the stratified folds), and the bootstrap intervals are conditional on the catalogue and classifiers (they do not include regenerating the EoS populations or retraining). *Observational realism.* The errors are independent and Gaussian; real NICER posteriors are correlated and non-Gaussian, and the GW170817 and NICER posteriors themselves were not used. *Not comparable to earlier data.* The new data come from a new solver and cannot be compared point by point with the single-sequence data set.

7 Conclusions

Evaluated on populations of EoS with whole-EoS hold-out, machine-learning classification of neutron-star composition is far less decisive than the 97% single-sequence benchmark suggested: the accuracy plateau is about 65% even without measurement error (49% in an alternative population with different parametrisations), it appears to be set by the information in the observables rather than by the learner, strange matter is the only scenario that is cleanly separable (in the family defined here), and no discrimination between nucleonic and dark-matter-admixed matter is demonstrated in these experiments when the nuclear EoS is unknown (the dark component shifts f by a median of 0.57% at fixed nuclear EoS, a shift that depends on the baryon-number prescription of the one-fluid model by a median of 5% (at most 6%) for an individual EoS inside the stability window and by up to 65% above it). With errors anchored on current X-ray measurements and a forecast f-mode precision the accuracy is 33–37%, and calibrated prediction sets contain about 3.22 of four classes.

Data, code and reproducibility

The single-sequence analysis uses `audit.py`, `repro.py`, `blocksize.py`, `main.py`, `extrap.py`, `interp_check.py` and `report.py`; the population study uses the solver and generator in `population/` (`eos.py`, `tov.py`, `fmode2.py`, `families.py`, `gen_pop.py`, `gen_base.py`, `paired_analysis.py`, `families_b.py`, `analytic_benchmarks.py`, `convergence_extra.py`, `dm_prescription.py`, `prior_width.py`, `robust_report2.p` `stat_robust.py`, `cross_family.py`, `frozen_test.py`, `published_tau_comparison.py`, `bh_qnm.py`, `robust_report.py`), the derivation scripts (`derive_vac2.py`, `derive_matter.py`, `elim.py`), the experiment (`pop_experiment.py`) and the report scripts. The code and data are on the `revised-analysis` branch of the repository below (the version described here is the release tagged `v2.0.0`, whose archived copy and DOI are to be given here before submission); the convergence tests are in `convergence_tests.py`. The 360 generated EoS of the primary population (plus the additional records of the wide-prior sensitivity run) and all results, including those of the wide-prior run (`results_pop_*_wide.*` and `wide_prior_results/`), are included as JSON/CSV; figures, tables and every number in the text are generated from them. Fixed seeds are used throughout. The environment is Python 3.13.0 with scikit-learn 1.7.2. The released single-sequence data are at `https://github.com/wasif553/ML-NeutronStar-Classification`.

A Single-sequence noise study (supplement)

This appendix keeps the noise, oracle and uncertainty analysis performed on the single-sequence benchmark, with blocked validation. It is superseded by the population study for questions about composition, but it isolates the effect of measurement error on the original data and provides an exact oracle bound.

Mass regimes, errors, learners, oracle

Three mass regimes are used: **A**, all models ($N = 770$); **B**, $M \geq 0.5\,M_\odot$ ($N = 395$, with 40 dark-matter models); **C**, $M \geq 1\,M_\odot$ ($N = 232$, three classes). Contiguous mass blocks of six models are assigned at random to training (60%), calibration (20%) and test (20%) roles; the analysis is repeated for 10 random assignments. Observables are perturbed as $x_j(1 + s\sigma_j\epsilon_j)$ with $(\sigma_M, \sigma_R, \sigma_f, \sigma_\tau) = (5, 5, 2, 20)\%$ at $s = 1$. The learner is a three-member MLP ensemble, with logistic regression and a random forest as baselines. The oracle Bayes classifier knows the exact sequences (including the test blocks) and the error model,

$$P(k \mid \mathbf{x}^{\mathrm{obs}}) \;\propto\; \frac{1}{N_k} \sum_{i \in k} \prod_j \mathcal{N}\Big(x_j^{\mathrm{obs}};\, x_{ij},\, (s\sigma_j x_{ij})^2\Big)\,, \tag{2}$$

and is an upper bound for learners trained on finite data.

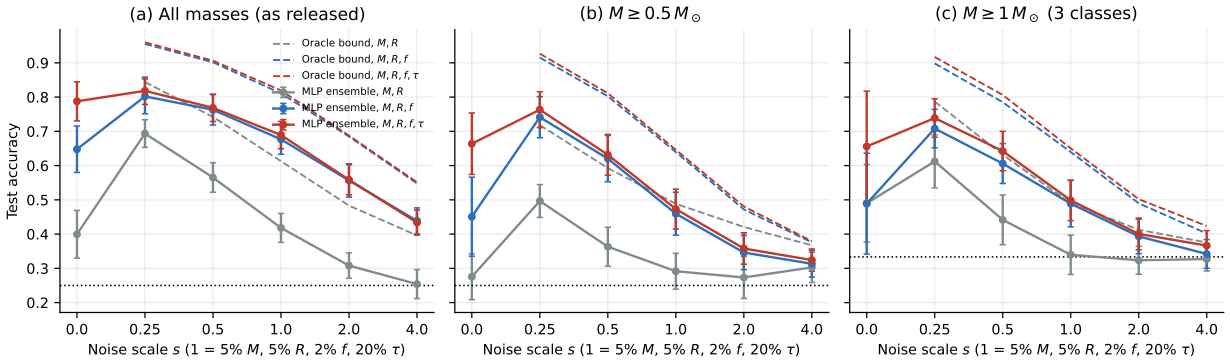


Figure 11: Single-sequence data: test accuracy versus noise scale s for the MLP ensemble (solid, 95% t-intervals over 10 repetitions) and the oracle Bayes bound (dashed). Dotted: chance.

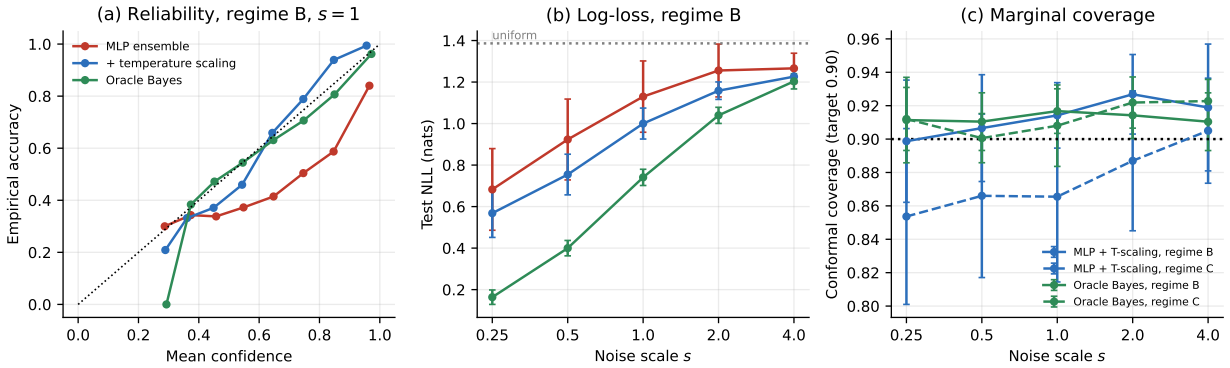


Figure 12: Single-sequence data: (a) reliability diagram (regime B, $s = 1$), (b) NLL, (c) conformal coverage versus noise scale.

Results

At $s = 1$ in regime B the ensemble reaches 47% with M, R, f, τ against an oracle ceiling of 65% (chance 25%); M and R alone give 29% (oracle 49%). In regime C (three classes) the values are 50% (ensemble) and 65% (oracle). Adding f raises the oracle from 49% to 64%; adding a 20% τ gives 65%. Strange and hyperonic matter are the most confused pair (36% of strange models labelled hyperonic and 37% the reverse). The raw ensemble is overconfident (ECE 0.188, NLL 1.13 at $s = 1$, regime B; oracle 0.040, 0.74); temperature scaling reduces them to 0.096 and 1.00. Conformal sets have about 90% marginal coverage and contain 2.35 of four classes at $s = 1$ (oracle 1.85). When the heaviest 25% of each class is withheld, the ensemble reaches 50% (oracle 80%) and the conformal sets cover only 0.81, since exchangeability fails; the ensemble's mutual information rises from 0.010 to 0.025 nats, an incomplete warning.

Table 13: Single-sequence data: test accuracy (%, mean ± 95% t-interval over 10 repetitions) by regime, observable set and noise scale s. Chance is 25% (A, B) or 33% (C).

Regime	Observables	s	MLP ensemble	Logistic reg.	Random forest	Oracle bound
A	M, R	0.25	69.4±4.0	35.9±5.4	21.3±3.5	84.4±2.7
A	M, R	1	41.8±4.2	37.1±4.5	24.8±3.5	61.3±4.2
A	M, R	4	25.4±4.2	29.1±3.7	23.1±2.5	39.6±4.1
A	M, R, f	0.25	80.2±5.1	38.7±4.9	56.0±4.9	95.5±1.1
A	M, R, f	1	67.7±4.4	38.7±5.2	56.3±3.5	81.1±2.8
A	M, R, f	4	43.9±3.8	38.3±4.3	39.5±2.7	54.8±3.3
A	M, R, f, τ	0.25	81.8±4.0	44.8±4.8	58.9±3.7	96.0±1.0
A	M, R, f, τ	1	68.9±4.0	40.3±4.3	59.6±3.8	81.9±2.8
A	M, R, f, τ	4	43.4±3.7	37.8±4.2	39.4±2.9	55.1±3.1
B	M, R	0.25	49.7±4.8	40.6±6.0	30.9±4.7	71.8±3.9
B	M, R	1	29.2±5.2	35.9±5.7	28.1±3.5	48.8±2.4
B	M, R	4	30.3±4.4	32.1±4.3	30.1±1.6	36.7±2.5
B	M, R, f	0.25	74.1±6.0	59.5±5.2	40.8±5.8	91.5±1.9
B	M, R, f	1	46.0±6.3	45.5±4.3	38.8±3.8	63.9±2.5
B	M, R, f	4	31.3±3.9	35.1±4.3	30.6±2.6	37.6±2.2
B	M, R, f, τ	0.25	76.3±5.2	63.6±3.9	43.4±5.7	92.7±1.8
B	M, R, f, τ	1	47.3±5.8	46.4±4.4	39.9±4.3	64.6±2.7
B	M, R, f, τ	4	32.4±3.2	35.4±4.2	31.8±2.2	37.9±2.1
C	M, R	0.25	61.2±7.7	62.6±7.5	37.7±8.7	78.7±4.2
C	M, R	1	34.0±5.7	40.7±7.1	33.1±5.5	49.0±4.8
C	M, R	4	32.7±3.5	33.9±6.2	33.2±2.5	37.5±5.8
C	M, R, f	0.25	70.8±5.6	69.0±6.7	44.1±10.0	89.8±1.6
C	M, R, f	1	48.9±6.8	51.4±6.6	41.9±6.7	64.0±3.5
C	M, R, f	4	34.2±4.2	36.0±5.1	34.3±3.4	40.2±4.3
C	M, R, f, τ	0.25	73.9±5.6	72.5±6.4	50.0±9.6	91.7±1.4
C	M, R, f, τ	1	49.9±5.9	52.5±7.0	44.1±6.9	65.0±3.1
C	M, R, f, τ	4	36.6±4.4	36.6±5.3	35.7±3.1	42.4±4.7

B Experimental settings

Table 14: Settings.

Item	Setting
Population folds	5 folds by EoS (stratified), test / calibration / training = 1 / 1 / 3 folds, 3 random partitions
Population samples	training: 33 masses × 3 noise draws per EoS; calibration: 33 × 2; test: 33 × 3 (single draw when noise-free)
MLP	3 members, 2×64 ReLU, Adam, lr 10^{-3}, $\alpha = 10^{-4}$, batch 128, early stopping (max 120 iterations)
Other learners	logistic regression (L2); k-NN, $k = 25$; random forest, 200 trees, min leaf 5; histogram gradient boosting (150 iterations); RBF SVM, $C = 3$, 8000 samples
Kernel Bayes	reference: 22 masses per EoS; training EoS, or all non-test EoS
Robustness studies	EoS-level bootstrap: 2000 resamples within class; learning curves: 25, 50, 75, 100% of the training EoS; population B: 48 EoS per scenario, same folds protocol; out-of-family: train on all EoS of one population, test on all of the other, three noise seeds
Numerical tests	integration tolerances 10^{-8}–10^{-13}, surface cut-off 10^{-6}–10^{-4}, centre radius ×0.5–2.5, matching radius 1500–8000 km, crust pressure scale ±10%
Conformal	LAC score, $\alpha = 0.1$; one star per calibration EoS (average of 40 random choices), quantile level $\lceil (n+1)(1-\alpha) \rceil / n$ with n the number of calibration EoS; pooled variant as sensitivity check
Single-sequence study	blocks of six models, 60/20/20, 10 repetitions; 20 noise draws per training model; random forest 150 trees; 15 repetitions for the block-length study, 100 for the random-split replication

C Changes relative to arXiv:2512.23758v3

Table 15: Statements of the earlier version that are withdrawn or changed.

Earlier statement	Status
Each class is one EoS; the classifier "separates composition"	Replaced by populations of EoS with whole-EoS hold-out; the accuracy ceiling is about 65% and no discrimination between nucleonic and dark-matter-admixed matter is demonstrated.
≈ 190 models per class, 770 in total	Total is correct; classes contain 222, 220, 122 and 206 models.
Models span $\sim 1\,M_\odot$ to the maximum mass	The data span 7×10^{-8} to $2.16\,M_\odot$; 70% have $M < 1\,M_\odot$.
97.4% test accuracy	Reproducible (97.7% average over 100 splits) but inflated by neighbouring models; falls to 94–69% with six- to 24-model blocks.
Per-class accuracy ranges 82–97% "across dataset realizations"	Not reproduced; removed.
"Mild degradation near the high-mass end"	Not supported by the earlier figure; removed.
"Not merely overfitting noise"	The inputs were noise-free; replaced by explicit error scenarios.
Permutation importance shows f and τ dominate	Replaced by observable-set ablations: f carries the information, τ adds nothing.
Tabulated redshift used as an input	Inconsistent with (M, R) for two classes; removed.
Oscillation quantities from the earlier code	Population data generated with a new solver, checked against six f-mode frequencies of an unpublished same-author cross-code pilot, analytic benchmarks and a published damping-time relation.

References

A. N. Angelopoulos and S. Bates, Found. Trends Mach. Learn. **16**, 494 (2023) [arXiv:2107.07511].

T. Damour and A. Nagar, Phys. Rev. D **80**, 084035 (2009).

S. Detweiler and L. Lindblom, Astrophys. J. **292**, 12 (1985).

Y. Fujimoto, K. Fukushima, K. Murase, Phys. Rev. D **98**, 023019 (2018).

B. Fornal and B. Grinstein, Phys. Rev. Lett. **120**, 191801 (2018).

T. Hinderer, Astrophys. J. **677**, 1216 (2008).

W. Husain, S. Motta, A. W. Thomas, arXiv:2203.02758 (2022).

W. Husain, oscillation (f-mode and gravitational-wave) calculations supplying f, τ, Q, h_c of the single-sequence data set; cited as "Husain (2026)" in arXiv:2512.23758v3 (full citation to be completed by the author).

W. Husain, *Testing f-mode information beyond static structure in neutron-decay dark-matter stars: a finite-EoS pilot and working research manuscript*, 15 October 2026 (unpublished working manuscript).

G. Lioutas and N. Stergioulas, Gen. Relativ. Gravit. **50**, 12 (2018) [arXiv:1709.10067].

E. W. Leaver, Proc. R. Soc. Lond. A **402**, 285 (1985).

B. Lakshminarayanan, A. Pritzel, C. Blundell, Adv. Neural Inf. Process. Syst. **30** (2017).

J. M. Lattimer and M. Prakash, Science **304**, 536 (2004).

M. C. Miller et al., Astrophys. J. Lett. **887**, L24 (2019).

J. R. Oppenheimer and G. M. Volkoff, Phys. Rev. **55**, 374 (1939).

F. Özel and P. Freire, Annu. Rev. Astron. Astrophys. **54**, 401 (2016).

S. Postnikov, M. Prakash, J. M. Lattimer, Phys. Rev. D **82**, 024016 (2010).

J. S. Read, B. D. Lackey, B. J. Owen, J. L. Friedman, Phys. Rev. D **79**, 124032 (2009).

T. E. Riley et al., Astrophys. J. Lett. **887**, L21 (2019).

J. Rikovska Stone, P. A. M. Guichon, H. H. Matevosyan, A. W. Thomas, Nucl. Phys. A **792**, 341 (2007).

R. C. Tolman, Phys. Rev. **55**, 364 (1939).

V. Vovk, A. Gammerman, G. Shafer, *Algorithmic Learning in a Random World* (Springer, 2005).

N. Williams, G. Pratten, P. Schmidt, Phys. Rev. D **105**, 123032 (2022) [arXiv:2203.00623].

F. J. Zerilli, Phys. Rev. D **2**, 2141 (1970).